INCH-POUND

MS27242L
21 October 2020
SUPERSEDING
MS27242K
27 November 2003

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 100 AMPERES, 1 PST N. O. TYPE I, HERMETICALLY SEALED

INACTIVE FOR NEW DESIGN AFTER 1 FEBRUARY 2000.
NO SUPERSEDING SPECIFICATION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

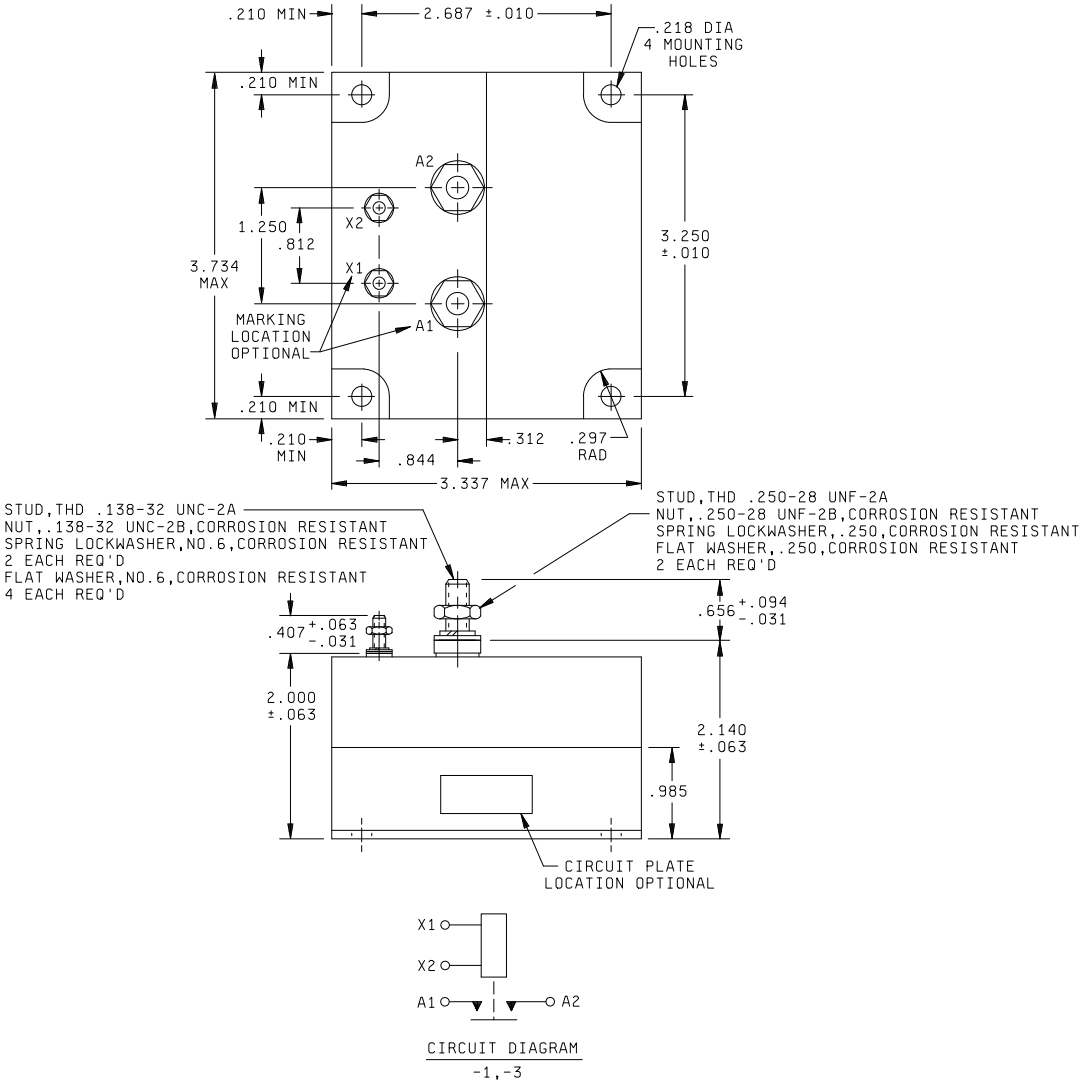


FIGURE 1. Dimensions and configuration.

AMSC N/A

FSC 5945

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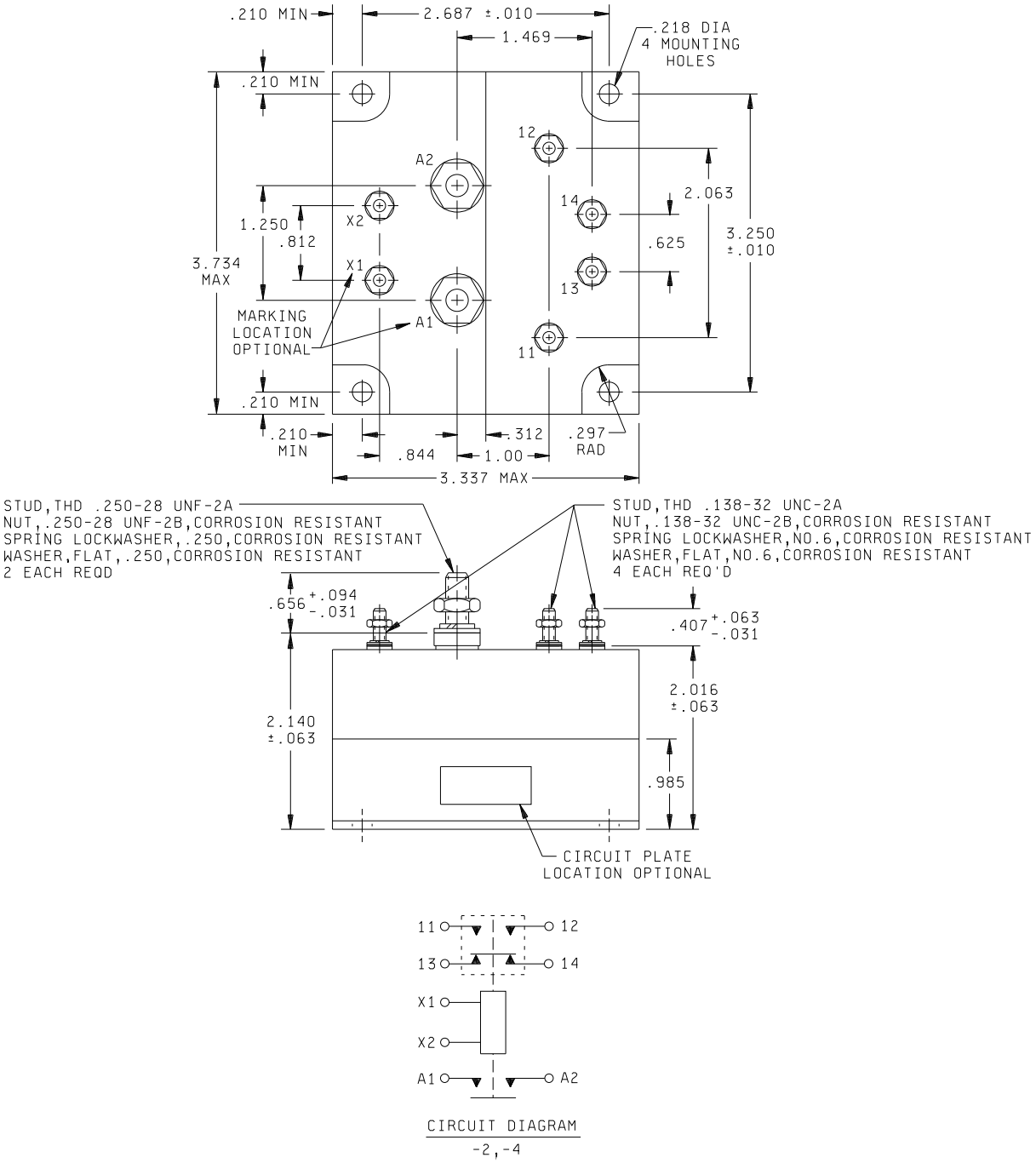


FIGURE 1. Dimensions and configuration - Continued.

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Inches	mm	Inches	mm	Inches	mm
.010	0.25	.407	10.34	2.063	52.40
.031	0.79	.625	15.88	2.140	54.36
.063	1.60	.656	16.66	2.687	68.25
.094	2.39	.812	20.62	3.250	82.55
.138	3.51	.844	21.44	3.337	84.76
.210	5.33	.985	25.01	3.734	94.84
.218	5.54	1.250	31.75		
.250	6.35	1.469	37.31		
.297	7.54	2.000	50.80		
.312	7.92	2.016	51.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. Coil and auxiliary terminals may use additional washer for terminal seat.
5. Terminal covers and barriers may be required at power or auxiliary terminals.
6. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
7. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this standard to the extent specified herein.
8. For detail information, see tables I through IV.

FIGURE 1. Dimensions and configurations - Continued.

TABLE I. Relay characteristics.

PIN number MS27242-	Type	Coil type	Terminal type	Mounting or mating socket	Auxiliary contacts	Max weight wt. lbs. 1/
1	I	dc	Stud	Flange	None	1.20
2	I	dc	Stud	Flange	Yes	1.25
3	I	ac	Stud	Flange	None	1.25
4	I	ac	Stud	Flange	Yes	1.30

1/ Weights include terminal barriers and covers.

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TABLE II. Operating characteristics.

PIN MS 27242-	Coil data												Time - (milliseconds maximum)					
	Coil	Rated			Max		Max pick-up voltage			Hold votl-age 3/	Drop out volt-age 3/	Oper -ate 1/	Re-lease 4/	Contact Bounce 1/			
														Main		Aux	
		Volts 2/	Freq Hz	Res Ω ±10%	Volts	Amp	Nor-mal 3/	High temp test	Cont cur-rent test					NO	NC	NO	NC
1	X1,X2	28	dc	51	29	.6	18	21	22.5	7.0	1.5	25	10	2	---	---	---
2	X1,X2	28	dc	51	29	.6	18	21	22.5	7.0	1.5	25	10	2	---	2	2
3	X1,X2	115	60/ 400	N/A	124	.2	90	100	104	40	10	25	50	2	---	---	---
4	X1,X2	115	60/ 400	N/A	124	.2	90	100	104	40	10	25	50	2	---	2	2

1/ With rated coil voltage.
2/ Caution: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
3/ Over the temperature range.
4/ From rated coil voltage.

Qualification by similarity: See MIL-PRF-6106.

Part or Identifying Number (PIN): MS27242- (plus dash number from table I).

TABLE III. Rated contact load (amperes per pole) (case grounded).

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	100	---	5	5	100	---	5	---	---	---	---	---	
Inductive	10	100	---	5	5	100	---	5	---	---	---	---	---	
Inductive														
Motor	50	100	---	---	---	---	---	---	---	---	---	---	---	
Lamp	---	---	---	1	1	---	---	1	---	---	---	---	---	
Transfer load														2/
Mechanical life reduced current	100	25	---	1.25	1.25	25	---	1.25	---	---	---	---	---	
Mixed loads	50	Applicable in accordance with MIL-PRF-6106												

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

TABLE IV. Vibration level.

PIN MS27742-	5 - 10 Hz	10 - 55 Hz	55 - 250 Hz	250-500 Hz	500-1500 Hz
1	.08 DA	.06 DA	15 g	15 g	12 g
2	.08 DA	.06 DA	10 g	10 g	10 g
3	.08 DA	.06 DA	15 g	15 g	15 g
4	.08 DA	.06 DA	10 g	10 g	10 g

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Environmental characteristics:	
Temperature range	-70°C to +125°C
Maximum altitude rating	80,000 ft
Shock G-level	50 g's
Duration	6-9 ms
Maximum duration contact opening	2 ms
Vibration - sinusoidal	(See table IV)
Vibration - random	N/A
Acceleration	15 g's
Electrical characteristics:	
Minimum insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms
Dielectric strength (sea level) (see table V)	2-5 seconds

Group B and Group C inspections are not required.

Table V. Dielectric strength (sea level).

Dielectric strength	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250	1,500	1,000	1,125
Aux contacts	1,250		1,000	
All other points	1,800		1,350	

Dielectric strength (altitude) (see table VI) 1 minute.

TABLE VI. Dielectric strength (altitude).

Dielectric strength	28 V dc	115 V ac
Coil to case	500	500
Aux contacts	500	500
All other points	700	700

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Qualification by similarity: See MIL-PRF-6106.

Maximum contact drop initial	0.150 volts
After life test	0.175 volts
Overload current (NO)	800 amperes
Rupture current (NO)	1,000 amperes
Duty rating	Continuous
RFI specification	MIL-STD-461

(Applicable to coil circuits of ac operated relays.)

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:

MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2020-062)

Review activity:
Navy – EC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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